Literature Review

Viewpoint Invariance as a Structural Guarantee for
Rehabilitation Exercise Assessment from Skeleton Sequences

Abstract

This document surveys the work our paper builds on and departs from. We organise it along the six axes the paper touches — (i) rehabilitation exercise assessment from skeletons, (ii) skeleton-based action recognition, (iii) group-equivariant and invariant architectures, (iv) learned versus structural viewpoint invariance, (v) continuous-time models for irregular sampling, and (vi) edge deployment of structured models — and after each we state the specific gap that motivates a contribution. A synthesis table (§8) maps every gap to the contribution that closes it. The through-line is a single distinction the existing literature does not make operational: a symmetry that holds *in expectation over a training distribution* (augmentation) versus one that holds *identically for every input* (architecture). Our thesis is that for a clinical screening instrument only the latter is admissible, and that it is provable, certifiable, and deployable.

1 Scope and Problem Statement

Automated assessment of physical rehabilitation exercises from skeleton sequences promises objective, remote, continuous monitoring of therapy [1]. A model receives a 3D skeleton trajectory — typically from a consumer depth sensor — and regresses a clinical quality score. The dominant evaluation regime ranks architectures by a single aggregate error against a physician's score on clean, frontal, single-sensor recordings [3, 4, 5, 2].

Our paper argues that this regime, not the architectures, is the limiting factor: it rewards capacity and penalises nothing a deployed home device actually encounters — a camera against a different wall, a tracking node that freezes, a dropped frame. The review below is therefore structured to expose two recurring blind spots in the prior art: (a) evaluation is uniformly *in-distribution*, and (b) no prior work on this task reports a *mean-predictor floor*, without which a robustness curve cannot be interpreted (a model that ignores its input has zero degradation and would "win" every robustness plot).

2 Rehabilitation Exercise Assessment from Skeletons

Benchmarks. KIMORE [1] is the standard corpus: subjects perform five trunk and hip exercises, each scored 0–50 by a clinician, with paired healthy and clinical populations. UI-PRMD [7] provides ten movements with marker-based ground truth; IRDS (the IntelliRehab dataset) contributes gesture-level rehabilitation recordings. Ismail-Fawaz et al.'s Rehab-Pile [6] aggregates KIMORE, UI-PRMD and IRDS under cross-subject splits and reports error rather than rank correlation — its protocol discipline is the closest antecedent of our own protocol audit.

Methods. Reported approaches span a wide architectural range: ST-GCN variants trained with supervised contrastive objectives using hard and soft negatives [3]; cross-modal video-to-joints augmentation to enlarge scarce training data [4]; dual-stream spatio-temporal graph networks with motion-aware grouping [5]; and, most relevant to us, point-cloud transformers that treat the skeleton as an unordered set of 3D points with axial self-attention [2]. The last is the baseline we position against directly.

What the field measures, and what it does not. Two properties recur. First, evaluation is uniformly in-distribution: same sensor, same camera placement, no simulated hardware failure. Second, protocols differ enough that headline numbers are not comparable — reported MADs on the most common single-exercise slice mix optimistic epoch-selection rules with subject-disjointness assumptions that do not hold uniformly.

Gap: No prior rehab-assessment work (a) stress-tests off the frontal, clean, single-sensor manifold, or (b) reports a mean-predictor floor against which a degradation curve is interpretable. Protocol incomparability is acknowledged anecdotally but never isolated quantitatively.

3 Skeleton-Based Action Recognition

The methodological backbone of skeleton scoring is the action-recognition literature. Spatial-temporal graph convolutional networks (ST-GCN) [8] established the graph-over-joints, convolution-over-time paradigm that most rehabilitation models inherit, and remain the standard baseline for cross-view and cross-subject generalisation on NTU RGB+D [9]. NTU RGB+D's *Cross-View* protocol — train on two physical camera setups, test on a third displaced by 45° — is the field's canonical *real* (non-synthetic) viewpoint-generalisation benchmark, and we use it precisely to show our viewpoint guarantee is not an artifact of synthetic test-time rotation.

Gap: Cross-view robustness in action recognition is pursued almost exclusively by augmentation or view canonicalisation, i.e. as a statistical property of a trained model over a view distribution — never as an exact, per-prediction architectural identity. Occlusion/tracking failure is treated as a nuisance to be masked away, not as a discriminating experiment between otherwise-equivalent architectures.

4 Equivariant and Invariant Neural Architectures

Group-equivariant networks impose symmetry as a *constraint on the hypothesis class* rather than a target of fitting [10]. For 3D geometry the steerable family — tensor-field networks [11], $SE(3)$-Transformers [12], and the `e3nn` framework [13] — carries features in irreducible representations of $O(3)$ and constrains every layer to intertwine with the Wigner-D action, guaranteeing equivariance *for arbitrary weights* (trained, untrained, or adversarial). Lighter constructions obtain $E(n)$-equivariance without spherical harmonics [14], or lift a network to $SO(3)$-equivariance by replacing scalar neurons with vector-valued ones [15]. An orthogonal line makes a non-equivariant backbone invariant by canonicalisation or frame averaging [16] — exact, but at a forward-pass or learned-canonicaliser cost.

These architectures are mature in molecular modelling and point-cloud vision but have not been brought to bear on rehabilitation scoring as an *exact viewpoint guarantee*. Two further points are, to our knowledge, unstated in this literature: (a) the natural invariant read-out one forms from such an encoder — built from norms, inner products, and bone lengths — is composed entirely of parity-*even* features and is therefore silently invariant to a *mirror* of the whole body, i.e. complete for $O(3)$

rather than $SO(3)$, and thus blind to trajectory handedness until pseudo-scalars are admitted; and (b) whether the steerable machinery is *necessary* for accuracy is rarely tested against a hand-crafted invariant baseline of equal guarantee.

Gap: Equivariant architectures give an exact viewpoint guarantee "for free" but (i) this has not been certified numerically for a rehabilitation read-out, (ii) its parity/chirality completeness is unexamined, and (iii) its necessity over a simple hand-crafted invariant model has not been isolated by a discriminating experiment.

5 Learned versus Structural Viewpoint Invariance

The default remedy for viewpoint variation is training-time augmentation, and it "works" in the sense the field measures: an augmented model's *aggregate* error becomes approximately flat in camera angle. But augmentation makes prediction errors *cancel* in the population mean rather than *vanish* per input. The distinction between a symmetry that holds in expectation over a training distribution and one that holds identically for every input is, for a clinical instrument, the whole point — a screening tool must return the *same patient the same score from a different room* — yet it is expressible only if degradation is reported *per sequence*, not as an aggregate.

Gap: The field reports aggregate accuracy versus camera angle, which a rotation-augmented model can flatten. It does not report per-sequence score shift, so a model whose errors merely cancel is indistinguishable from one that is genuinely invariant.

6 Continuous-Time Models and Irregular Sampling

Depth-sensor streams are not uniformly sampled — genuine multi-frame dropouts and jittered inter-arrival times are the norm. The standard treatment resamples onto a uniform grid; the standard alternative is a continuous-time model consuming arrival times natively: Neural ODEs [17], latent ODEs and ODE-RNNs for irregular series [18], Neural CDEs driven by an interpolant of the observations [19], and recurrences with explicit decay between observations (GRU-D) [20], all built on gated recurrent units [21]. The widely assumed benefit is that native consumption of irregular timestamps must beat resampling.

Two subtleties in this literature are load-bearing for us. First, adaptive solvers such as Dormand–Prince [22] accept or reject steps on a *scalar* reduction of the embedded local error; if that scalar is not group-invariant, a rotated and an unrotated integration receive *different step grids* and silently de-align — even when the vector field is exactly equivariant and the representation orthogonal. Second, irregularly sampled data have no single Nyquist frequency and do not alias, which is the basis of the Lomb–Scargle construction [23]: a grid-free estimator can recover a high-frequency tone that a uniform grid at the same *average* rate folds onto itself irrecoverably.

Gap: The advantage of native irregular-time consumption is assumed, not conditioned on the signal's spectral content; and the interaction between adaptive-solver error control and group equivariance is undocumented, so a "certified equivariant" continuous-time model can still silently break invariance through step-grid divergence.

7 Edge Deployment of Structured Models

A guarantee proved in exact (64-bit) arithmetic on a full recording is not the same as a guarantee that survives on the hardware a home device runs: reduced precision (fp16/bf16/int8) and a real-time latency budget. Quantisation is not an orthogonal transformation, so it does not commute with the group action, and a high-accuracy skeleton classifier that is bidirectional and pools over the whole clip cannot emit a score until the last frame — it has no honest latency.

Gap: The precision budget of an equivariance guarantee (where, exactly, the theorem degrades from fp64 to int8, and whether it is weight- or activation-quantisation that breaks it) is unmeasured; and viewpoint-robust skeleton scorers are not co-designed for strictly causal, frame-by-frame real-time operation.

8 Synthesis: Gaps and How This Work Closes Them

Table 1 collects the gaps above and maps each to the contribution that addresses it. The unifying claim is that clean-accuracy leaderboards cannot separate architectures on this task, and that the decisive variable is *structural behaviour off the clean, frontal manifold*.

9 Positioning of Our Novelty

Relative to the literature above, the paper's novelty is not a new equivariant layer or a new benchmark score. It is a *reframing of what should be measured* on this task, plus the structural machinery and certification to act on it:

1. **Assessment as an invariant read-out of an equivariant representation.** We formulate rehabilitation scoring so that invariance of the predicted score under any rigid motion of the camera is a *theorem about the architecture*, true for arbitrary weights, rather than a behaviour induced by augmentation — and we *certify* it numerically (E1–E8) rather than assert it.

2. **The measurement discipline the task was missing.** A mean-predictor floor, a quantified nondeterminism floor (≈ 0.33 MAD) below which no accuracy gap is real, and per-sequence degradation reported alongside aggregate accuracy everywhere. This turns "robustness" from an uninterpretable plot into a gated claim.

3. **Two negative results, one of them derived.** We report — rather than hide — that native irregular-time consumption buys nothing on KIMORE, and we *predict* the null from a spectral census before running the experiment; and that restoring chirality is principled but unrewarded on a corpus of healthy-form exercises. Honest scope boundaries, with the corpus properties that would reward each capability named explicitly.

4. **A discriminating experiment, not an appeal to elegance.** Because a hand-crafted invariant GRU matches the steerable encoder on both accuracy and the viewpoint guarantee, we justify the steerable machinery by the one axis on which they measurably differ — graceful degradation under a frozen tracking node — and we record, against ourselves, that the point-cloud transformer is in fact the most node-robust arm, so the defensible claim is a Pareto one.

5. **The guarantee as a deployable systems property.** A structural quantisation audit (weight- vs. activation-rounding) and a strictly causal streaming read-out show the theorem is co-designed for the edge, not merely an fp64 identity.

In one sentence: prior work optimises clean, frontal, aggregate accuracy; we show that on this task that quantity cannot separate architectures, and that the decisive, clinically relevant variable — a per-prediction viewpoint guarantee that survives real cameras, a broken sensor, reduced precision, and real-time latency — is *structural*, provable, and measurable.

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Gap in prior work	How this work closes it
No mean-predictor floor; protocol incomparability unquantified	Protocol audit isolates optimistic epoch selection as a 16.8% effect on the standard KIMORE slice, shows that slice carries no subject-level signal, fixes a pooled subject-disjoint protocol, and gates every claim on clearing the floor.
Exact viewpoint invariance never certified for a rehab read-out	An $SE(3)$-invariant read-out over a steerable encoder [13], audited by eight numerical gates (E1–E8): orthogonality $\leq 10^{-12}$, violation at machine roundoff with a *flat* slope in rotation angle, an 8×10^8 fp32/fp64 ratio; cross-viewpoint degradation 9×10^{-6} MAD vs. the baseline's 9.42.
Viewpoint robustness only shown on synthetic rotation	Confirmed on NTU RGB+D 60 Cross-View [9, 8], a *real* 45° cross-camera benchmark, at exact (5.8×10^{-9}, fp64) cross-view logit invariance without rotation augmentation.
Augmentation's cost reported only in aggregate	The rotation-augmented baseline is granted its remedy and its price measured: a flat +1.0 MAD penalty *and* a 3.03 MAD per-sequence score shift that the aggregate hides — a flat accuracy curve is shown to *not* be invariance.
Parity/chirality completeness of invariant cuts unexamined	The natural invariant cut is exhibited (gate E6) to be silently $O(3)$-invariant and blind to trajectory handedness; pseudo-scalars restore $SO(3)$ at no cost to the viewpoint theorem (E8).
Necessity of steerable machinery not isolated	A GRU on hand-crafted $SO(3)$ invariants matches accuracy *and* the exact guarantee at a third of the parameters; the steerable encoder is then justified by a *measured* sensor-node-failure experiment, not rhetoric.
Irregular-sampling advantage assumed, not conditioned	Stated as a bandwidth condition and verified in both directions: a spectral census shows resampling destroys only 2.2% of KIMORE's positional energy (a denoiser), predicting the null; a 5 Hz tremor placed in the destroyed band is recovered at $r = 1.000$ under 70% drops via a Lomb–Scargle-style estimator [23].
Solver error control vs. equivariance undocumented	The per-component RMS norm of a Dormand–Prince controller [22] is proved non-invariant; a per-irrep-normalised norm drives step-grid divergence to identically zero. The Neural CDE [19] is certified to the same standard yet *fails* the floor — a clean demonstration that a structural guarantee is worthless without measured signal.
Precision budget and real-time latency of the guarantee unmeasured	The equivariance floor is measured fp64→int8; weight-quantisation preserves the theorem exactly while only activation-rounding breaks it; the read-out streams causally at 8.9 ms/frame (112 fps) with an honest time-to-first-score.

Table 1: Gap-to-contribution map. Each row is a shortfall identified in §2–§7 and the measured result that addresses it.